Systematic Search for Singularities in 3D Euler-Voigt Flows

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Agenda

Singularities & Euler-Voigt Flows

- Navier-Stokes system ($\Omega = [0, L]^3$)

$$\begin{cases} \partial_t \mathbf{u} + (\mathbf{u} \cdot \boldsymbol{\nabla})\mathbf{u} + \boldsymbol{\nabla} p - \nu \Delta \mathbf{u} = \mathbf{0}, & \text{in } \Omega \times (0, T] \\ \boldsymbol{\nabla} \cdot \mathbf{u} = 0, & \text{in } \Omega \times (0, T] \\ \mathbf{u} = \boldsymbol{\eta} & \text{in } \Omega \text{ at } t = 0 \\ \text{Periodic Boundary Conditions} & \text{on } \partial\Omega \times (0, T] \end{cases}$$

- **The Big Question:**

  Given a smooth initial condition η, does the Navier-Stokes system always admit smooth solutions $\mathbf{u}(t)$ for arbitrarily long times t?

- One of the Clay Institute "Millennium Problems" ($ 1M prize!)

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- $\alpha > 0$ is the regularization parameter with units of length. Acts as a 'spectral filter'. The Euler-Voigt system is globally well-posed given smooth initial conditions (Y. Cao et al. 2006).
- As $\alpha \to 0$ the solutions of Euler-Voigt converge to the solutions of Euler and so its behaviour can be used to determine if the Euler equations will develop a finite-time singularity.

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- Proven by Theorem 1 in (A. Larios et al. 2018).

Theorem

Let $\boldsymbol{\eta}_\alpha \in H^s$, $s \geq 3$ with $\boldsymbol{\nabla} \cdot \boldsymbol{\eta}_\alpha = 0$ and let $\boldsymbol{u}_\alpha$ be the corresponding unique solution of Euler-Voigt. Suppose that there exists a time $T^* > 0$ such that

$$\limsup_{\alpha \to 0^+} \left(\alpha \sup_{t \in [0, T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2} \right) > 0. \tag{1}$$

Then the Euler equations with the initial condition $\boldsymbol{\eta}_\alpha$ must develop a singularity within the interval $[0, T^*]$.

- Theorem 5.1 (A. Larios et al. 2010).

Theorem

Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ *for some* $s \geq 3$*, let*

$$\mathbf{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\mathbb{T}^3)) \cap C^1([0,T], H^{s-1}(\mathbb{T}^3))$$

be the corresponding solutions to Euler and Euler-Voigt, respectively, where $0 < T < T_{max}$ *and* T_{max} *is the maximal time for which a solution to Euler exists and is unique. For all* $t \in [0,T]$*, we have*

$$\begin{aligned}&\|\mathbf{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\mathbf{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \leq \\ &\left(\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2\right) e^{Ct} + C\alpha^2 (e^{Ct} - 1).\end{aligned}$$

- Modified from Theorem 5.2 (A. Larios et al. 2010).

Theorem

Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\Omega)$ for some $s \geq 3$ satisfying

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad \text{as } \alpha \to 0,$$

let $\mathbf{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\Omega)) \cap C^1([0,T], H^{s-1}(\Omega))$ be the corresponding solutions to Euler and Euler-Voigt, respectively. If there exists a finite time $T^* > 0$ such that the solutions $\boldsymbol{u}_\alpha$ satisfies

$$\limsup_{\alpha \to 0} \left(\alpha^2 \sup_{t \in [0, T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) > 0,$$

then $\mathbf{u}$ develops a singularity in the interval $[0, T^*]$.

Proof.

Suppose that the solution $\mathbf{u}$ of the Euler equations Euler with the initial condition η is regular on $[0, T^*]$. Since the Euler-Voigt equations are globally well-posed, we have

$$\|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{\eta}_\alpha\|_{L^2}^2,$$

thus

$$\alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{\eta}_\alpha\|_{L^2}^2 - \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2.$$

Taking the supremum over $t \in [0, T^*]$ of the above equation, we obtain that

$$\sup_{t\in[0,T^*]} \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2.$$

Proof (Cont.)

Taking the limsup as $\alpha \to 0$, we obtain that

$$\begin{aligned}\limsup_{\alpha\to 0}\left(\sup_{t\in[0,T^*]}\alpha^2\|\boldsymbol{\nabla}\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right) &= \limsup_{\alpha\to 0}\left(\|\boldsymbol{\eta}_\alpha\|_{L^2}^2+\alpha^2\|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2-\inf_{t\in[0,T^*]}\|\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right)\\ &= \|\boldsymbol{\eta}\|_{L^2}^2-\|be\|_{L^2}^2\\ &= 0\end{aligned}$$

which contradicts the hypothesis in the theorem that the above expression is positive.

Proof (Cont.)

The second equality holds because

$$\begin{aligned}
\limsup_{\alpha\to 0}\left|\inf_{t\in[0,T^*]}\|\boldsymbol{u}_\alpha(t)\|_{L^2}-\|\boldsymbol{\eta}\|_{L^2}\right| &\le \limsup_{\alpha\to 0}\|\boldsymbol{u}_\alpha(t_\alpha)-\boldsymbol{\eta}\|_{L^2}^2\\
&\le \limsup_{\alpha\to 0}\|\boldsymbol{u}_\alpha(t_\alpha)-\mathbf{u}(t_\alpha)\|_{L^2}^2+\|\mathbf{u}(t_\alpha)-\boldsymbol{\eta}\|_{L^2}^2\\
&= 0.
\end{aligned}$$

Here we use the fact that the convergence of the solutions of the Euler-Voigt equations to the solution of the Euler equations is uniform on $[0, T^*]$, as shown in Theorem 5.1. □

- We consider analytic initial conditions that belong to the following Gevrey class

$$G^\sigma := \left\{ \mathbf{v} \in H^s(\Omega) : \|\mathbf{v}\|_{G^\sigma} = \langle \mathbf{v}, \mathbf{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0.$$

- We select the following inner product

$$\forall \mathbf{v}, \mathbf{u} \in G^\sigma, \quad \langle \mathbf{v}, \mathbf{u} \rangle_{G^\sigma} := \sum_{\mathbf{j} \in \mathbb{Z}^3} (1 + |2\pi \mathbf{j}|^2)^s e^{4\pi\sigma|\mathbf{j}|} \hat{\mathbf{v}}_{\mathbf{j}} \cdot \bar{\hat{\mathbf{u}}}_{\mathbf{j}}$$
$$= \int_\Omega (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v} \cdot \mathbf{u} \, d\mathbf{x}.$$
$$= \left\langle (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v}, \mathbf{u} \right\rangle_{L^2}$$

- The operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[\widehat{|D| \boldsymbol{v}} \right]_{\boldsymbol{j}} := 2\pi |\boldsymbol{j}| \hat{\boldsymbol{v}}_{\boldsymbol{j}}, \qquad \left[\widehat{e^{\sigma|D|} \boldsymbol{v}} \right]_{\boldsymbol{j}} := e^{2\pi|\boldsymbol{j}|} \hat{\boldsymbol{v}}_{\boldsymbol{j}}.$$

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- We assume zero mean velocity without loss of generality.

$$V := \{\boldsymbol{v} \in G^{\sigma} : \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0, \quad \int_{\Omega} \boldsymbol{v}\, d\mathbf{x} = 0\}.$$

- We use a closed manifold $\mathcal{M}_1 \subset V$ defined as

$$\mathcal{M}_1 = \{\mathbf{v} \in V : \|\mathbf{v}\|_{\dot{H}^1} = C\}.$$

- Euler ($\mathbf{u}$) and Euler-Voigt ($\boldsymbol{u}_{\alpha}$) time scaling property

$$\tilde{\mathbf{u}}_{(\alpha)}(\mathbf{x}, t) = \lambda \mathbf{u}_{(\alpha)}(\mathbf{x}, \lambda t), \tag{6}$$

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- Taylor Green Vortex ($\Omega = [0, L]^3$)

$$\eta_{TGV} = \begin{bmatrix} V_0 \sin(2\pi x/L)\cos(2\pi y/L)\cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L)\sin(2\pi y/L)\cos(2\pi z/L) \\ 0 \end{bmatrix}$$

- Characteristic time-scale

$$t_c = \frac{L}{V_0}$$

- Using ($\Omega = [0, 2\pi]^3$) and $V_0 = 1$, Bustamente et a. (2012) showed numerical evidence for finite-time singularity in Euler from η_{TGV} at $T^* \approx 4$. We use ($\Omega = [0, 1]^3$) and so scale down to $V_0 = 1/2\pi$.

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- Calculate C analytically

$$C = \|\boldsymbol{\eta}\|_{\dot{H}^1} = \|\nabla\boldsymbol{\eta}\|_{L^2} = \left[\sum_{i,j=1}^{3}\int_\Omega (\partial_i \eta_j)^2\,dx\right]^{1/2}$$

- Calculating for scaled η_{TGV}

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- Beale-Kato-Majda (BKM) criterion (Beale et al. 1984): A Smooth solution u of the Euler system develops a singularity at $t = t_0$ if and only if

$$\lim_{t \to t_0} \int_0^t \|\omega\|_{L^\infty} d\tau = \infty, \qquad \boldsymbol{\omega} = \nabla \times \mathbf{u}$$

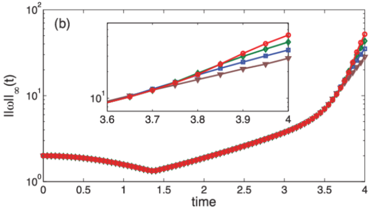

Figure: (Bustamente et al. 2012 Fig. 1b) Maximum of vorticity $\|\omega(\cdot, t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 512^3 (brown triangles), 1024^3 (blue squares), 2048^3 (green diamonds), and 4096^3 (red circles).

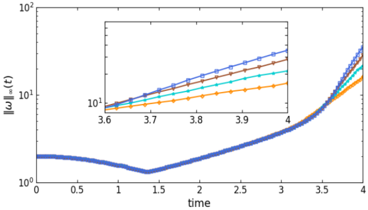

Figure: Maximum of vorticity $\|\omega(\cdot, t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), 1024^3 (blue squares).

Optimization Problem for Euler-Voigt

- Define our objective function $\Phi_{\alpha,T} : V \to \mathbb{R}^+$

$$\Phi_{\alpha,T}(\boldsymbol{\eta}) := \|\nabla \boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)\|_{L^2}^2 = \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)\|_{\dot{H}^1}^2$$

- Given $\alpha, T \in \mathbb{R}^+$, find

$$\tilde{\boldsymbol{\eta}}_{\alpha,T} = \underset{\boldsymbol{\eta}_\alpha \in \mathcal{M}_1}{\operatorname{argmax}} \, \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha).$$

- If $\alpha^2 \Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T}) > 0$ as $\alpha \to 0$ then a singularity forms in Euler the interval $[0, T]$.

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- The first order finite difference equation for the Gâteaux differential $\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}'_\alpha) : V \times V \to \mathbb{R}$,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}'_\alpha) := \lim_{\epsilon\to 0} \frac{1}{\epsilon}\left[\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha + \epsilon\boldsymbol{\eta}'_\alpha) - \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)\right].$$

- Substituting perturbations into Euler-Voigt

$$\boldsymbol{u}_\alpha \to \boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha, \eta_\alpha \to \eta_\alpha + \eta'_\alpha, \text{ and } p \to p + p'$$

- Linearization of Euler-Voigt

$$\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha \\ p'\end{bmatrix} := \begin{bmatrix}\partial_t \boldsymbol{u}'_\alpha + (I - \alpha^2\Delta)^{-1}(\boldsymbol{u}_\alpha \cdot \nabla \boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha \cdot \nabla \boldsymbol{u}_\alpha + \nabla p') = 0 \\ \nabla \cdot \boldsymbol{u}'_\alpha\end{bmatrix} = \begin{bmatrix}0 \\ 0\end{bmatrix}$$
$$\boldsymbol{u}'_\alpha(x,0) = \eta'_\alpha$$

- The first order finite difference equation for the Gâteaux differential $\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) : V \times V \to \mathbb{R}$,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) := \lim_{\epsilon \to 0} \frac{1}{\epsilon} \left[\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha + \epsilon \boldsymbol{\eta}'_\alpha) - \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) \right].$$

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- Integration by parts on the following duality-pairing relation

$$\begin{aligned}\left(\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha \\ p'\end{bmatrix}, \begin{bmatrix}\boldsymbol{u}^*_\alpha \\ p^*\end{bmatrix}\right) &:= \int_0^T \int_{\mathbb{T}^3} \mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha \\ p'\end{bmatrix} \cdot \begin{bmatrix}\boldsymbol{u}^*_\alpha \\ p^*\end{bmatrix} \, d\mathbf{x}\, dt \\ &= \left(\begin{bmatrix}\boldsymbol{u}'_\alpha \\ p'\end{bmatrix}, \mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha \\ p^*\end{bmatrix}\right) + \int_{\mathbb{T}^3} \boldsymbol{u}'_\alpha(\mathbf{x}, T) \cdot \boldsymbol{u}^*_\alpha(\mathbf{x}, T)\, d\mathbf{x} \\ &\quad - \int_{\mathbb{T}^3} \eta'(\mathbf{x}) \cdot \boldsymbol{u}^*_\alpha(\mathbf{x}, 0)\, d\mathbf{x} \\ &= 0.\end{aligned}$$

- Adjoint System

$$\mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha \\ p^*_\alpha\end{bmatrix} := \begin{bmatrix}-\partial_t \boldsymbol{u}^*_\alpha - (\nabla(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha + (\nabla(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha)^\intercal)\boldsymbol{u}_\alpha - \nabla p^*_\alpha \\ -\nabla \cdot \boldsymbol{u}^*_\alpha\end{bmatrix} = \begin{bmatrix}0 \\ 0\end{bmatrix}$$

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- Substituting $\Phi_{\alpha,T}$ into the Gâteaux differential,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha(\mathbf{x},T), \boldsymbol{u}'_\alpha(\mathbf{x},T)\right\rangle_{\dot{H}^1} = 2\left\langle \boldsymbol{u}^*_\alpha(\mathbf{x},0), \boldsymbol{u}'_\alpha(\mathbf{x},0)\right\rangle^2_{L^2}$$

- From the Riesz representation theorem (Luenberger 1969)

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}:\boldsymbol{\eta}') = \left\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha), \boldsymbol{\eta}'_\alpha\right\rangle_{G^\sigma}$$

- The gradient of the objective functional is

$$\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) = (1+|D|^2)^{-3}e^{-2\sigma|D|}\boldsymbol{u}^*_\alpha(x,0)$$

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- Tangent space to $\mathcal{M}_1$ at $\mathbf{v}$

$$\mathcal{T}_{\mathbf{v}}\mathcal{M}_1 = \left\{ z \in V : \langle \mathbf{v}, z \rangle_{\dot{H}^1} = 0 \right\}$$

- For any $z \in \mathcal{T}_{\mathbf{v}}\mathcal{M}_1$

$$\begin{aligned}
\langle \mathbf{v}, z \rangle_{\dot{H}^1} &= \int_{\mathbb{T}^3} |D|\mathbf{v} \cdot |D|\mathbf{z}\, d\mathbf{x} \\
&= \int_{\mathbb{T}^3} (1+|D|^2)^3 e^{2\sigma|D|} \left((1+|D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v} \right) \cdot \mathbf{z}\, d\mathbf{x} \\
&= \left\langle (1+|D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}, \mathbf{z} \right\rangle_{G^\sigma} = 0
\end{aligned}$$

- Characterized using the unit 'normal' vector

$$\mathbf{n}_{\mathbf{v}} = \frac{(1+|D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}}{\|(1+|D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}\|_{G^\sigma}}$$

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- Projection operator $P_{\mathcal{T}_{\mathbf{v}}\mathcal{M}_1} : V \to \mathcal{T}_{\mathbf{v}}\mathcal{M}_1$

$$\forall \mathbf{w} \in V, \qquad P_{\mathcal{T}_{\mathbf{v}}\mathcal{M}_1}\mathbf{w} = \mathbf{w} - \langle \mathbf{n}_{\mathbf{v}}, \mathbf{w} \rangle_{G^\sigma} \mathbf{n}_v$$

- Retraction operator $R : V \to \mathcal{M}_1$

$$R(\mathbf{v}) = \frac{\sqrt{3}}{2}\frac{\mathbf{v}}{\|\mathbf{v}\|_{\dot{H}^1}}, \qquad \mathbf{v} \neq 0$$

- Vector field ξ transported along $\mathcal{M}_1$ in the direction of ϕ

$$\mathcal{T}\mathcal{M}_1 \oplus \mathcal{T}\mathcal{M}_1 \to \mathcal{T}\mathcal{M}_1 : (\varphi, \xi) \longmapsto \Gamma_\varphi(\xi) \in \mathcal{T}\mathcal{M}_1,$$

where $\mathcal{T}\mathcal{M}_1 = \cup_{\mathbf{v}\in\mathcal{M}_1}\mathcal{T}_{\mathbf{v}}\mathcal{M}_1$ is the tangent bundle on $\mathcal{M}_1$ (absil et al. 2008). It is calculated via the differentiated retraction

$$\Gamma_\varphi(\xi) = \frac{d}{dt}R(\varphi + t\xi)\Big|_{t=0}.$$

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- We can now construct the iterative relation

$$\boldsymbol{\eta}_\alpha^{(n+1)} = R\left[\boldsymbol{\eta}_\alpha^{(n)} + \tau_n \mathbf{d}^{(n)}\right]$$

- Let $\boldsymbol{G}^{(n)} := \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^{(n)})$. Let $\mathcal{T}_n := \mathcal{T}_{\boldsymbol{\eta}_\alpha^{(n)}}\mathcal{M}_1$. $\mathbf{d}^{(n)}$ is computed as

$$\begin{aligned}\mathbf{d}^{(0)} &= P_{\mathcal{T}_0}\boldsymbol{G}^{(0)} \\ \mathbf{d}^{(n)} &= P_{\mathcal{T}_n}\boldsymbol{G}^{(n)} + \beta_n \Gamma_{\tau_{n-1}\mathbf{d}^{(n-1)}}(\mathbf{d}^{(n-1)}), \quad n \geq 1.\end{aligned}$$

- 'Momentum' term is computed using the Polak-Ribière approach (Nocedal et al. 1999).

$$\beta_n = \frac{\left\langle P_{\mathcal{T}_n}\boldsymbol{G}^{(n)}, \left(P_{\mathcal{T}_n}\boldsymbol{G}^{(n)} - \Gamma_{\tau_{n-1}\mathbf{d}^{n-1}} P_{\mathcal{T}_{n-1}}\boldsymbol{G}^{(n-1)}\right)\right\rangle_{G^\sigma}}{\|P_{\mathcal{T}_{n-1}}\boldsymbol{G}^{(n-1)}\|_{G^\sigma}^2}$$

- Step size

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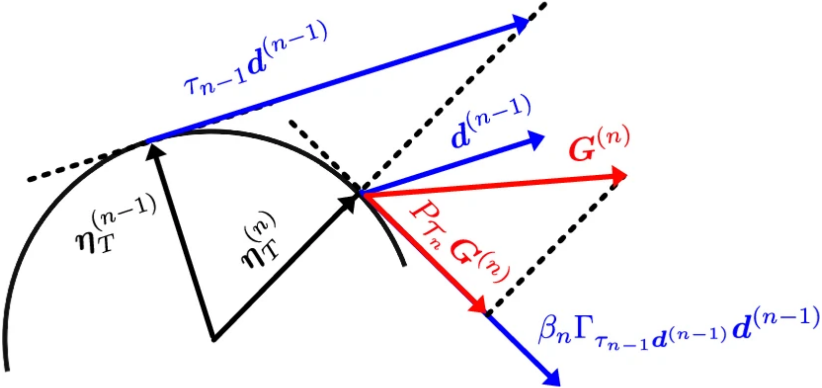

Figure: (Zhao et al. 2023, Fig. 1)

- Periodic algorithm restart ($\beta_n = 0$)
  - $$n = 20k, \quad k \in \mathbb{Z}^+$$
  - The search direction $\mathbf{d}^{(n)}$ is not an ascent direction
    $$\frac{\left\langle \mathbf{d}^{(n)}, P_{\mathcal{T}_n} \boldsymbol{G}^{(n)} \right\rangle_{G^\sigma}}{\|\mathbf{d}^{(n)}\|_{G^\sigma} \|P_{\mathcal{T}_n} \boldsymbol{G}^{(n)}\|_{G^\sigma}} < \epsilon, \quad 0 < \epsilon \ll 1.$$
- Iterations are stopped when the relative change of $\Phi_{\alpha,T}$ is below a specified tolerance
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- Discretized in space using a standard pseudospectral Fourier method
- Derivatives are evaluated in the Fourier space
- Nonlinear products are computed in the physical space on an uniform grid with N points in each direction.
- Explicit fourth-order Runge-Kutta
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- Domain

$$\text{resolution} : 128^3, 256^3, 512^3, 1024^3$$
$$\alpha : 1/1024, 2/1024, 4/1024, 8/1024, 16/1024$$

- Time

$$dt : 2^{-5}$$
$$T : 3, 5$$

- Gevrey class

$$\sigma : \text{1E-5, 1E-4, 1E-3, 1E-2, 1E-1}$$
$$s : 3$$

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- Time

$$dt : 2^{-5}$$
$$T : 3, 5$$

- Gevrey class

$$\sigma : \text{1E-5, 1E-4, 1E-3, 1E-2, 1E-1}$$
$$s : 3$$

- Domain

$$\text{resolution} : 128^3, 256^3, 512^3, 1024^3$$
$$\alpha : 1/1024, 2/1024, 4/1024, 8/1024, 16/1024$$

- Time

$$dt : 2^{-5}$$
$$T : 3, 5$$

- Gevrey class

$$\sigma : \text{1E-5, 1E-4, 1E-3, 1E-2, 1E-1}$$
$$s : 3$$

Results

- Evolution of $\|\boldsymbol{u}_\alpha(\mathbf{x}, t)\|^2_{\dot{H}^1}$ over time for $\alpha = 16/1024$.

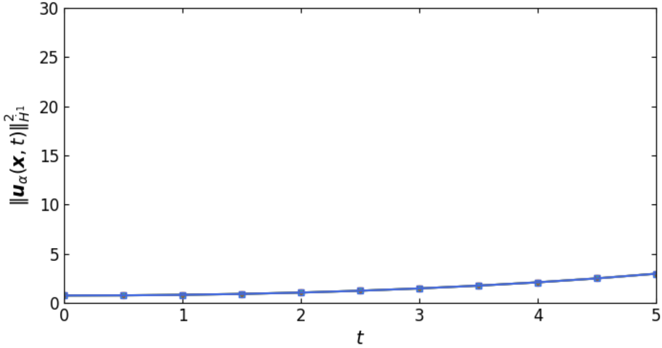

Figure: Results for $\alpha = 16/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\boldsymbol{u}_\alpha(\mathbf{x}, t)\|^2_{\dot{H}^1}$ over time for $\alpha = 8/1024$.

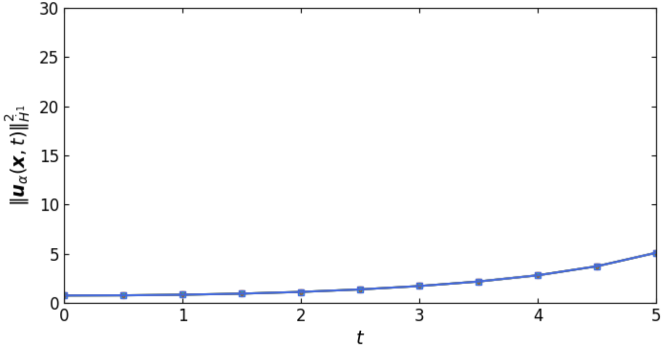

Figure: Results for $\alpha = 8/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\boldsymbol{u}_\alpha(\mathbf{x}, t)\|^2_{\dot{H}^1}$ over time for $\alpha = 4/1024$.

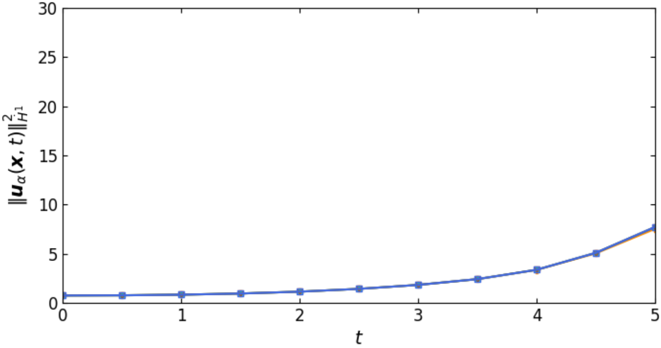

Figure: Results for $\alpha = 4/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\boldsymbol{u}_\alpha(\mathbf{x}, t)\|^2_{\dot{H}^1}$ over time for $\alpha = 2/1024$.

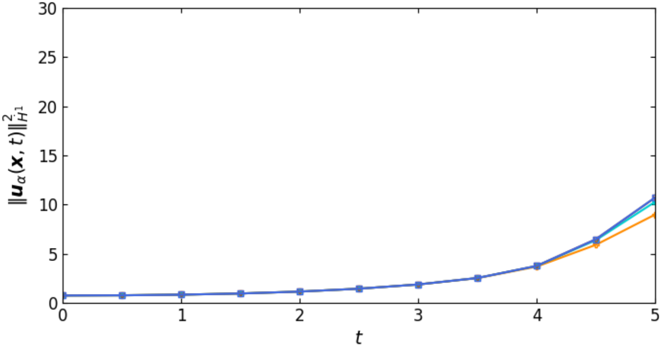

Figure: Results for $\alpha = 2/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\boldsymbol{u}_\alpha(\mathbf{x}, t)\|^2_{\dot{H}^1}$ over time for $\alpha = 1/1024$.

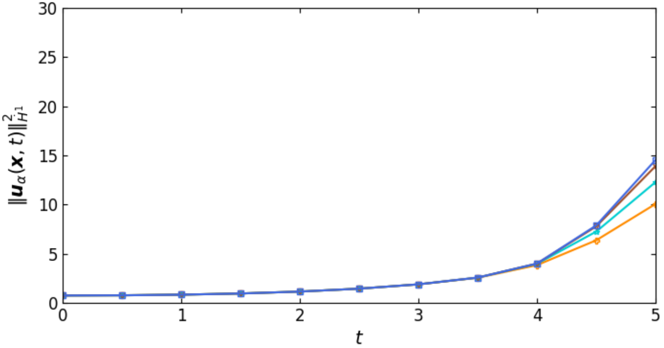

Figure: Results for $\alpha = 1/1024$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Evolution of $\|\mathbf{u}(\mathbf{x}, t)\|_{\dot{H}^1}^2$ over time for $\alpha = 0$.

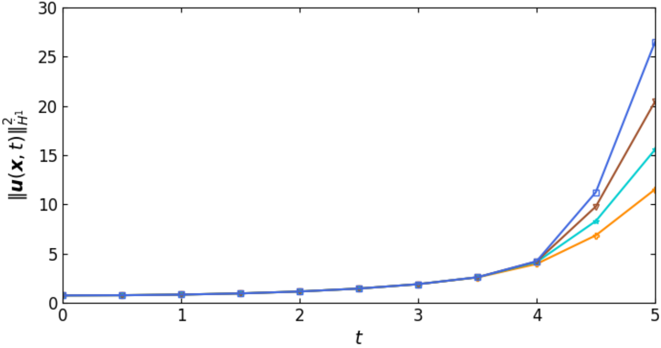

Figure: Results for $\alpha = 0$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$ for each resolution at $T = 3, 5$.

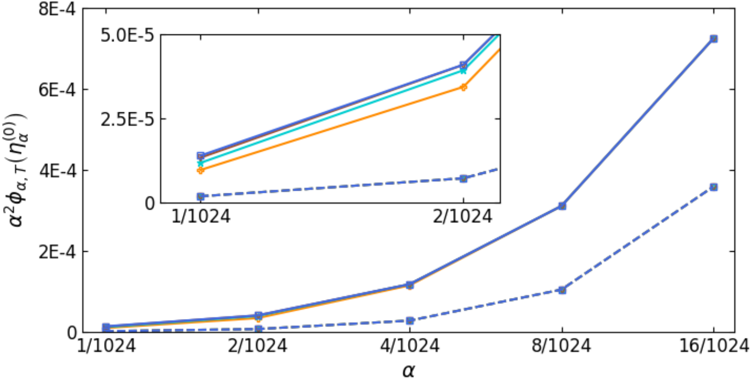

Figure: Results for $T = 3$ (dotted line) and for $T = 5$ (solid line). Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).